Programming Domains :

Scientific:
Large number of floating point computations
- Fortran (still used)
- ALGOL 60

Business:
Produce reports, use decimal numbers and characters
- COBOL

Artificial intelligence:
Symbols , consisting of names rather than numbers, are manipulated
- LISP
- Scheme
- Prolog

System programming :
systems SW(OS, support tools), must be efficient because of continuous use
- C, UNIX OS written in C

Web Software:
- markup, XHTML
- scripting, PHP
- general purpose, JavaScript

Language Evaluation Criteria:

1- **Readability**: the ease with which programs can be read and understood, it most Important measure of the quality of programs mainly because if maintenance and cost

Simplicity:
- a manageable set of features and constructs :

large number of basic constructs more difficult to learn
- feature multiplicity:

having more than one way to do an operation, ex: increment an integer variable in java
- operator overloading:

an operator symbol that has more than one meaning, ex: '+' used for both integer and floating point addition

Note: too much simplicity makes program less readable, ex: assembly

Orthogonality:
language constructs should not behave differently in different contexts, and the context of use should not cause unexpected restrictions or behavior

Most orthogonal language: Algol 68
- trade-off between orthogonality and simplicity is a central decision in language design

Note: too much orthogonality can cause a problems

Control statement:
goto statement severely reduces readability

Ex: early version of Fortran

Data types and structures:
the presence of meaningful data types aids readability

Ex: variable = 1 (unclear, no Boolean type in the language)
variable = true (clear, Boolean data type in the language)

Ex2: record data type to provide a method to represent variable records

Syntax design:
- Identifier length : affect readability flexible composition, it would be better if there is no restriction on the length
- Special words & methods of compound statements:

ex: difficult to determine which group is ended when an "end " or "}" appears

Note: Ada language uses "end if" and "end loop" to terminate compound construct (to make which group is ended more clear)
- Form and meaning(semantic):

ex: meaning of static in C depends on the context of its appearance

ex2: UNIX syntax does not suggest semantic (must be descriptive constructs & meaningful keywords)

2- Writability: the ease with which a language can be create programs

All characteristics that affected in **readability**

Support for abstraction:
The ability to define and use complex structures or operation on ways that allow details to be ignored, ex: sub program

Expressivity:
Set of relatively convenient ways to specifying operations
Ex: count++ better than count = count +1
Ex2: inclusion of for statement

Note: is not reasonable to compare writability of two languages when one was designed for that application, ex: VB for writing GUI program while C for writing system program, such as OS

3- Reliability: performance of a program to its specifications under all conditions

All characteristics that affected in **readability and writability:**

- The easier a program to write, the more likely it is to be correct
- Readability affects realizability in the writing and maintenance phases of the life cycle
- Programs difficult to read are difficult to write and modify

Type checking:
Testing for type errors, because run-time type checking is expensive, compile-time type checking (as in Java) is more desirable

Exception handling:
Intercept run-time errors and take corrective measures
Ex: Java, C++, Ada

Aliasing:
Having two or more distinct names to access the same memory cell
Ex: two pointers point to the same variable

4- Cost:

- **Training programming to use language**
- **Writing programs**
- **Compiling programs**
- **Executing programs:**
  Trade-off between compilation cost & execution speed
  Note:
  Optimization: collection of techniques that compiler may use to decrease the size or increase the execution speed
  > Increase the execution speed : use the optimization
  > Increase the compilation speed: doesn't use the optimization
- **language implementation system:**
  Availability of free compilers
  Ex: Java
- **Reliability:**
  Poor reliability leads to high costs
- **Maintaining programs**

Note: most important contributors to language cost are: development, maintenance, and reliability

5- Others:

- **Portability:**
  The ease with which programs can be moved from one implementation to another
- **Generality:**
  The applicability to wide range of applications
- **Well- define:**
  The completeness and precision of the language's official definition

Computer Architecture Influence

Computer architecture:

Most languages are developed around well-known computer architecture (Von Neumann architecture)

- Imperative languages: dominant because of Von Neumann computers

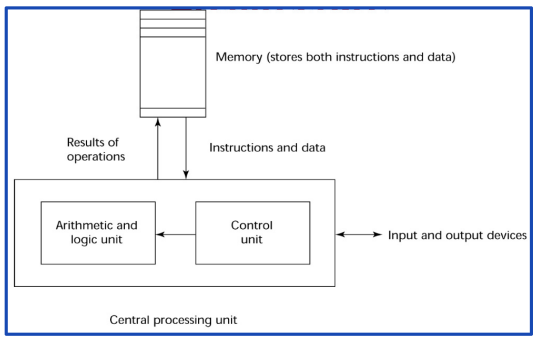
> Concept of Von Neumann :

1- Data and programs stored in memory

2- CPU is separate from memory

3- Instructions and data are transmitted (piped) from memory to CPU & results of operations in CPU moved back to memory

4- Basis (features) : variables model memory cells, assignment statement model piping, repetition), and iteration is efficient



- Fetch-execute-cycle on Van Neumann :

```
initialize the program counter (address of next instruction)
repeat forever
      fetch the instruction pointed by the counter
      increment the counter
      decode the instruction
      execute the instruction
end repeat
```

Programming design methodologies :

- Procedure-oriented programming: the opposite of data-oriented programming, and its methods have not been abended
- Data-oriented programming: focus on data design & abstract data types, and its methods dominate SW development

فيه معلومات زيادة في سلايد ٢٦ ، كاتبة عند التفصيل تبعها انه بس قراءة ما بيجي، بس مدري عني سحبت عليه :)

Languages Categories:

Imperative :

Features:

- Variables: model memory cell
- Assignment statement: based on piping operation
- Iteration: fast & efficient because instructions stored in adjacent memory cells
- Discourage use of recursion for repetition

Ex: C, Pascal, Java, C++

Functional:

Main means of computation is by applying functions to given parameters

Ex: LISP, Scheme

> Logical:

Rule-based

Ex: Prolog

Markup:

Used to specify the layout of information in Web documents

Ex: XHTML, XML

Object- oriented:

Data instruction, inheritance, dynamic binding

Ex: Java, C++

Language Design Trade-Offs: conflicting languages evaluation criteria leads to it

Reliability vs excursion cost :

Ex:

Java (unlike C) , demands all references to array elements be checked to ensure legal ranges, but that increased execution cost, so designer of Java traded execution efficiency for relibialty

Java —> more reliable

C—> faster

Writability (flexibility) vs Reliability :

Ex:

C++ pointers are powerful and very flexible but not reliable, so designer of C++ traded reliability for writability

Readability vs writability :

Ex:

APL offers many array operators > allowing complex computations in small program at the cost of readability

Note: Choosing constructs and features when designing a programming language requires many compromises and trade-offs

Programming Environment :

UNIX:
An older operating system programming environment and tool collection,
Used through a GUI that runs on top of UNIX

Borland JBluilder:
An integerated development environment for Java

Microsoft Visual Studio.NET:
A large visual environment, used to program in C#, Visual BASIC.NET, Jscript, J#, C++

Languages and environment Summary

ALGOL:
- Scientific (60)
- Most orthogonal (68)

Fortran:
- Scientific
- Early version of it use goto statement

Scheme:
- Artificial intelligence
- Functional

LISP:
- Artificial intelligence
- Functional

C++:
- Imperative
- Object-oriented
- Designer traded of reliability for writability
- Microsoft Visual Studio.NET

JavaScript:
- Web software (general-purpose)

Visual BASIC.NET:
Microsoft Visual Studio.NET

VB:
- For writing GUI program

XHTML:
- Web software (Markup)

C#:
Microsoft Visual Studio.NET

J#:
Microsoft Visual Studio.NET

PHP:
- Web software (scripting)

COBOL:
- Business
- Use compilers

Pascal:
- Imperative

Perl:
- Example for Hybrid implementation system

Prolog:
- Artificial intelligence
- Logic

APL:
- Designer traded readability for writability
- Example for pure interpretation

UNIX:
- Syntax does not suggest semantic
- Older operating system programming environment and tool collection, used through a GUI

Ada:
- Use end if and end loop
- Use compilers

C:
- Written UNIX OS
- Use compilers
- Syntax suggest semantic
- Imperative
- For writing program, such as OS
- Support aliasing
- Designer traded reliability for execution cost

XML:
- Web software (Markup)

Java:
- Imperative
- Object-oriented
- Widely used because of its free compiler
- Designer traded execution efficiency for reliability
- Example for hybrid implementation system
- Borland JBuilder

Assembly:
- Model of simplicity